

VWO: Optimizing web conversion rates with easy-to-use A/B testing solutions

Tell us your challenge. We're here to help.

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[About VWO](#)



VWO performed a
migration with zero
downtime, achieved a
fold boost to data
processing power
VWO latency by 4x
slashed carbon footprint
30% with Google

About VWO

With a mission to develop the world's best tools for optimizing a website, VWO helps global companies build extraordinary digital experiences with a SaaS solution that requires no tech background to use. It enables businesses to improve their website conversions using A/B testing, wherein two versions of a webpage are pitted against each other. Its founder Paras Chopra was featured in [Forbes list of 30 under 30](#)

(<https://www.forbes.com/pictures/efjl45ggme/paras-chopra/#1c5a519c3f9d>)

Indian and Asian tech innovators.

Google Cloud results

- Enables modern data warehouse creation with half the services required by competing providers, saving time and cost
- Keeps customers happy by enabling full-stack migration to Google Kubernetes Engine with no service interruptions
- Lowers VWO response time latency by 40% with Cloud Load Balancing
- Enables environmental ethos with

Smart
mark
insigl
30x in
data
powe

Industries: Technology

Location: India

(<https://cloud.google.com/load-balancing/>)



About Niveus Solutions

With more than 10 years of proven expertise in technical consultation, Niveus Solutions is a Google Cloud Partner that enables Indian innovators from healthcare to media and entertainment to optimize their digital transformation. It has executed infrastructure modernization on Google Cloud for 100+ customers and also specialize in custom build applications.



Google Cloud (<https://cloud.google.com/>)

ABOUT NIVEUS
Cloud AutoML (<https://cloud.google.com/automl/>)
SOLUTIONS

BigQuery (<https://cloud.google.com/bigquery/>)

Cloud Bigtable (<https://cloud.google.com/bigtable/>)

Cloud Load Balancing
(<https://cloud.google.com/load-balancing/>)

enables Indian innovators from healthcare
Cloud Storage (<https://cloud.google.com/storage/>)

Dataflow (<https://cloud.google.com/dataflow/>)
executed infrastructure modernization on

**30% reduction
in carbon
footprint**

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

Cloud Pub/Sub (<https://cloud.google.com/pubsub/>)

When Paras Chopra founded Wingify (<https://wingify.com/>) (the company behind VWO) in 2010, his goal was to make INR 50K in the first month, equivalent to \$1,000 at the time. He set that goal to prove to himself that his startup experiment could be a success and shared the first version of its flagship solution, VWO, on Hacker News (<https://news.ycombinator.com/>), a social news website for coders. A decade later, Wingify and VWO (<https://vwo.com/>) have an annual revenue of more than \$20 million, serving corporations in more than 90 countries around the world.

The journey was enabled by Paras' vision of building an A/B testing system so easy to use that marketers with no tech background could test new website features without the help of an engineering team. In A/B testing, two versions of a webpage are tested on visitors to deduce which one has a better impact on conversions. Empowering marketers with a plug-and-play tool to carry it out brings benefits of both creativity and efficiency.

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—Ankit Jain, Vice
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Wingify

VWO's vision is to build a comprehensive experimentation platform, providing a 360-degree picture of how to improve conversion rates based on A/B testing and visitor behavior insights. In order to succeed, VWO needed to transform its data

warehouse from bare-metal infrastructure to a cloud solution, crunching unlimited data with machine learning (ML) capabilities.

VWO turned to [BigQuery](#) (/bigquery) and its related [Google Cloud](https://cloud.google.com/) (https://cloud.google.com/) data solutions, tapping Google Partner [Niveus Solutions](https://niveussolutions.com/) (https://niveussolutions.com/) for expert insight in the transformation. The startup was so impressed that it decided to perform a complete "lift-and-modernize" migration to a containerized environment on [Google Kubernetes Engine](#) (/kubernetes-engine).

"We're a group of high-energy people who want to show the world that the best can come out of anywhere," says Ankit Jain, Vice President of Engineering at VWO. "When we discovered the power of Google Cloud, in particular BigQuery and Google Kubernetes Engine, we found solutions to transform the next phase of our evolution into reality."

"Google Cloud
delivered our
ideal data
warehouse
solution with
BigQuery,

Pub/Sub,
Dataflow, Cloud
Bigtable, and
Cloud Storage ...
It equated to a
smaller team,
spending fewer
hours, dealing
with a lot less
complexity. This
combination of
simplicity and
data analytics
expertise won us
over."

—Ankit Jain, Vice
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Wingify

Finding the right cloud partner for experimentation platform transformation with high data performance and proactive support

Ankit says the startup takes an extremely careful approach to all major operational decisions. That's why it asked Google Cloud and other cloud providers to offer competing proof-of-concept (POC) pitches

for the new data warehouse that would transform VWO's experimentation platform.

After spending weeks studying each proposal and observing the data analysis performance of the POCs in real time, VWO tapped the Google Cloud solution. The main reason for this was superior performance driven by a deeper base of expertise in data analytics and ML. Specifically, says Ankit, Google Cloud's data warehouse POC provided more robust data crunching using half of the components required by its competitors.

"Google Cloud delivered our ideal data warehouse solution with BigQuery, Pub/Sub (/pubsub), Dataflow (/dataflow), Cloud Bigtable (/bigtable), and Cloud Storage (/storage). That's five components compared to 10 used by other cloud providers," said Ankit. "It equated to a smaller team, spending fewer hours, dealing with a lot less complexity. This combination of simplicity and data analytics expertise won us over."

Another factor that stood out was the Google Cloud team itself, says Ankit. According to him, the team offered proactive, collaborative support at an engineer-to-engineer level, instead of the salesperson-to-customer relationship that he had found with other providers.

"The Google Cloud team wasn't focused on the end sale, but came to us as fellow engineers trying to understand our problems and work out solutions," says Ankit. He explains that the Google Cloud team sent product managers who were building systems

themselves as engineers, so they knew exactly what the issues were. This assured VWO that they were in good hands.

Providing unlimited data-processing power to enable full-fledged platform transformation

Ultimately, the BigQuery-led data warehouse brings VWO the unlimited data-processing power it requires to reinvent itself as a full-fledged experimentation platform, offering extensive analytics on the impact of experiments on various business metrics that help brands understand what their customers want.

From gigabytes of data per day, VWO today handles terabytes of data per day, a more than 1,000-fold boost to processing power. This has enabled it to open up its solution for clients to send any quantity of events to analyze for marketing experiments. Apart from the minimal cost, VWO is also able to serve its clients needs without worrying about infrastructure.



Wingify's #million2billion celebration in 2016 as they hit a million dollars in monthly recurring revenue.

Zero platform downtime during "lift-and-modernize" migration on Google Kubernetes Engine

VWO's positive experience with BigQuery inspired it to seize "a golden opportunity" for full-fledged modernization from physical servers to a containerized environment on Google Kubernetes Engine (GKE).

The data processing power of BigQuery and related services is complemented by unlimited (and granular) computing scalability from GKE. Since VWO was already moving its data warehouse, it took the opportunity to also migrate to a containerized environment, one it believed was fast becoming a tech industry standard. The move to GKE was also an intuitive one, because Kubernetes is developed by Google, which means it would be up to date on the latest Kubernetes updates.

For a global platform, the project posed a key challenge. How to migrate a daunting array of systems (from databases to platform code) without creating maintenance windows that would leave some customers from more than 90 countries without service? "If we're only serving customers in India, we could potentially carry out system upgrades in the middle of the night, and nobody would notice," says Ankit. "But since we're global, we felt we needed to remain connected around the clock so none of our clients would suffer interruptions."

VWO achieved that objective through a live incremental migration that leveraged Google Cloud's sprawling network of data centers. The strategy, devised in collaboration with Niveus Solutions, was to link VWO's US servers to Google Cloud's US East data center in North Virginia, only a millisecond of latency away.

After planning with Niveus and the Google Cloud team, the team carried out a piecemeal migration of databases, code, and other assets over 45 days. The latency advantage enabled by the data center network of Google Cloud meant that the migration could take place without any disruption of service to its global clients. Google Cloud solutions also enabled VWO to help clients with low latency in other key ways, for example, to ensure that its A/B testing SmartCode, which lives on client websites during marketing experiments, didn't slow down their websites due to remote connections.

Since VWO's clients are flung around the world, all website events must link to the closest server in VWO's global network to avoid latency issues. From Asia to Europe and North and South America, VWO has servers running in 12 regions on Google Cloud.

Cloud Load Balancing (/load-balancing) optimizes VWO's dynamic content delivery network (CDN) to ensure that client website visitor requests hit the nearest VWO server on the Google Cloud network in any region of the world. By deploying this CDN solution, it has been able to save around 40% in response times.

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Reducing carbon footprint with a shared vision of digital solutions

For VWO, one of the most rewarding performance metrics since moving to Google Cloud has been the drastic reduction in carbon footprint. One reason VWO turned to Google Cloud was its own commitment to being a 100% renewable energy solution (/sustainability), powering digital innovation while protecting the planet. This is in line with its founder Paras' commitment to the environment, demonstrated through climate change films (<https://wingify.earth>) it has commissioned and work done by Wingify Foundation (<https://wingifyfoundation.org/>), a project dedicated to rural sustainability projects in India.

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"It brings us great satisfaction to be building an innovative experimentation platform for clients around the world, while being friendly to the planet."